# PhD Research Plan

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| **Roll No.** | 23318006 |
| **Title of Research Work** | Design and Development of Artificial Intelligence (AI) Integrated UAV Platforms for Landmine Detection |
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This document presents an overview of PhD Research Plan.

## 1 Introduction

Landmines and unexploded ordnance (UXO) continue to pose a severe threat to human safety, especially in post-conflict regions. Traditional detection techniques such as manual probing, trained dogs, and ground vehicles equipped with sensors are effective but slow, labor-intensive, and dangerous. In recent years, Unmanned Aerial Vehicles (UAVs) have emerged as a transformative platform for remote sensing and surveillance in hazardous environments. Their capability to fly autonomously, cover large areas, and operate in challenging terrains makes them a promising tool for landmine detection applications [1].

Thermal imaging has become a powerful non-contact sensing method for detecting buried landmines. Buried objects including anti-personnel and anti-tank mines alter the thermal characteristics of the surrounding soil due to differences in heat absorption, insulation, compaction, moisture retention, and diurnal temperature cycling. These variations create subtle thermal anomalies on the surface, which can be captured by UAV-mounted thermal cameras with high spatial resolution. As a result, thermal imaging enables rapid detection of potential mine locations without requiring ground contact or physical disturbance. Figure 1 illustrates a UAV equipped with a thermal imaging module used for aerial scanning.

However, UAV-mounted thermal sensors introduce several challenges, including thermal noise from sunlight, soil temperature variability, vegetation cover, environmental humidity, surface texture differences, and platform vibrations. Detection performance also varies depending on the time of day typically early morning or late evening due to stronger thermal contrast between mines and surrounding soil. To address these limitations, researchers focus on optimizing flight altitude, thermal calibration routines, anomaly enhancement techniques, and robust image-processing pipelines.

Artificial Intelligence (AI) has significantly improved the reliability of thermal-based landmine detection. Deep learning and machine learning algorithms help automate anomaly recognition, reduce false positives, and enhance classification accuracy from thermal images [2]. Techniques such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs) have shown promise in identifying temperature-based patterns associated with buried mines, even under noisy conditions [3].

Simulation and modeling play a critical role in developing UAV-based thermal detection systems. Through simulated environments, researchers can study thermal propagation patterns, evaluate sensor response under different soil types and environmental conditions, and generate synthetic training data [4]. Tools such as MATLAB/Simulink and ROS/Gazebo are widely used to model UAV flight dynamics, sensor behavior, and AI-based detection frameworks [5]. These simulation platforms help reduce experimental risk and cost while accelerating algorithm development.

In summary, a UAV platform equipped with thermal imaging and supported by AI-driven interpretation represents an advanced and efficient solution for landmine detection. The combination of thermal anomaly analysis, autonomous aerial scanning, and intelligent pattern recognition forms the foundation for next-generation humanitarian demining systems.

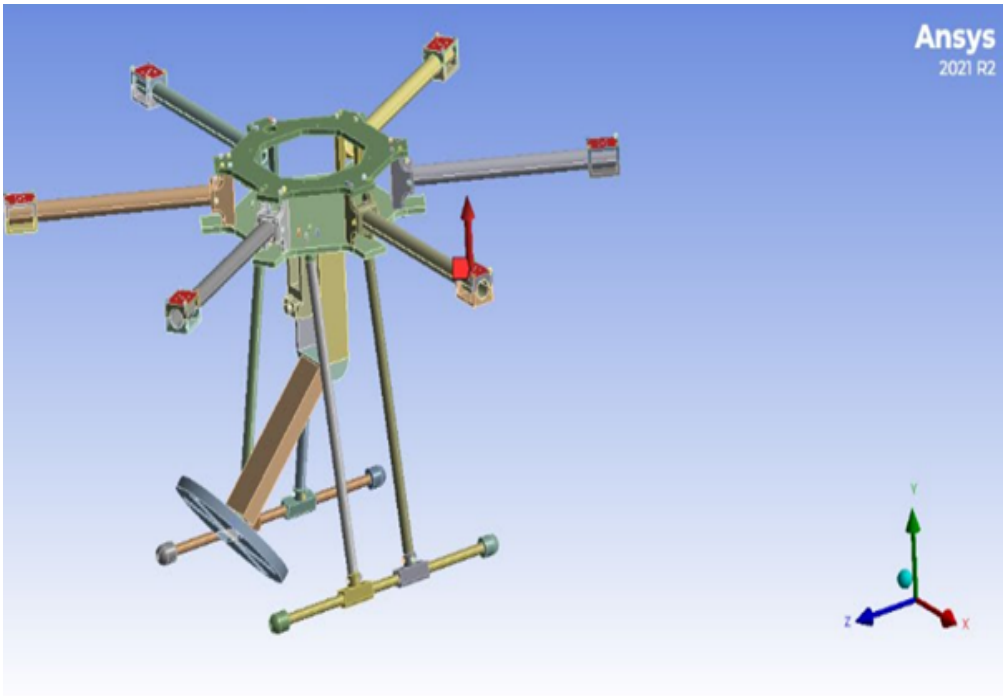



Figure 1: Drone integrated metal detection system

### 1.1 Advantages

The integration of Unmanned Aerial Vehicles (UAVs) with thermal imaging, enhanced through Artificial Intelligence (AI), provides an advanced and safe approach to landmine detection. The foremost advantage of this system is improved safety, as UAVs allow remote inspection of mine-contaminated regions, reducing direct human involvement in hazardous environments [1]. UAVs also offer exceptional mobility and accessibility, capable of surveying vast, remote, or uneven terrains that may be inaccessible or dangerous for ground-based teams [2].

Thermal imaging serves as the primary sensing modality, detecting temperature anomalies caused by disturbed soil, variations in heat retention, moisture differences, or thermal inertia around buried landmines. These anomalies become visible due to diurnal heating and cooling cycles, enabling the UAV to identify potential buried threats without physical contact. Compared

with traditional methods, thermal cameras enable rapid surface scanning, wide-area coverage, and non-intrusive detection in both humanitarian and military demining contexts.

The incorporation of AI-based image processing and pattern recognition further strengthens detection capability. Machine learning and deep learning algorithms particularly convolutional neural networks (CNNs) can automatically analyze thermal images, suppress background noise, enhance subtle temperature contrasts, and classify potential mine signatures with high precision [4]. Real-time or near-real-time processing enabled by onboard or edge computing systems allows immediate field assessment and decision-making, significantly improving operational responsiveness [5].

Additionally, simulation tools such as MATLAB/Simulink and ROS/Gazebo provide virtual testing environments where UAV flight parameters, thermal sensor behavior, and AI-based detection algorithms can be optimized before real-world deployment. Such simulation-driven development reduces risk, cost, and field-testing overhead. UAV-based systems are also highly scalable, enabling coordinated multi-drone missions for rapid large-area mapping an essential capability in post-conflict or humanitarian demining operations [7].

### 1.2 Disadvantages

Despite their numerous advantages, AI-integrated UAV platforms for landmine detection also face several technical and operational challenges. A primary limitation arises from the sensitivity of thermal imaging to environmental conditions. Thermal patterns vary significantly with soil moisture, time of day, vegetation cover, shading, and ambient temperature, which may obscure or distort heat-based anomalies associated with buried mines. Unlike controlled laboratory conditions, real-world terrain introduces fluctuations that reduce thermal contrast and complicate automated interpretation.

Another key challenge is power and payload limitation. Thermal cameras with high spatial resolution, along with onboard AI processing units, significantly increase power consumption, reducing flight duration and operational efficiency [6]. Furthermore, achieving stable and vibration-free thermal imaging requires careful UAV design and active stabilization mechanisms, as motion blur and temperature drift can lead to misclassification or missed targets.

Thermal imaging also demands precise timing: early-morning and late-evening windows typically offer the highest thermal contrast, limiting available operational hours. High winds, cloud cover, and rapid weather shifts further degrade sensor reliability. Processing large volumes of thermal data requires substantial computational resources [7], increasing system complexity and cost.

Additionally, UAV operations are constrained by regulatory restrictions related to airspace permissions, operational altitude, and flights over sensitive or populated areas [8]. A major research challenge is the scarcity of large, labeled thermal datasets of buried mines. This limits AI model generalization across seasons, soil types, and climatic conditions, reducing reliability when deployed in diverse environments.

In conclusion, although AI-integrated UAV systems equipped with thermal imaging represent a major advancement in safe and intelligent landmine detection, limitations related to environmental sensitivity, endurance, sensor calibration, and dataset scarcity must be addressed through improved hardware design, adaptive AI models, and rigorous field validation to ensure scalable and dependable real-world deployment.

### 1.3 Types of Landmines

#### 1.3.1 Anti-Personnel Blast Mines (APMs)

Engineered primarily to incapacitate or injure individuals, anti-personnel blast mines are typically emplaced at shallow depths to maximize their effectiveness against human targets.

- PMN series
- **VS-50**

#### 1.3.2 Minimum-Metal / Plastic Mines

These devices incorporate minimal quantities of metal or are predominantly constructed from plastic, thereby significantly reducing their observability with conventional metal detectors.

- Type 72B
- PFM-1 (“Butterfly” mine)

#### 1.3.3 Anti-Tank Blast Mines (ATMs)

Anti-tank blast mines contain comparatively large explosive charges and are designed to disable or destroy armored vehicles and other military platforms.

- TM-62
- M15

#### 1.3.4 Fragmentation Mines (High-Metal Content)

Fragmentation mines are characterised by the incorporation of pre-formed steel balls or other metallic fragments intended to produce a high-velocity fragmentation field upon detonation.

- M18 Claymore
- MON-50

## 2 Literature Review

Recent developments in artificial intelligence (AI), unmanned aerial vehicles (UAVs), and advanced remote sensing technologies have significantly accelerated progress in automated landmine detection. While traditional research has relied on subsurface sensing such as Ground Penetrating Radar (GPR), recent studies increasingly emphasize thermal imaging as a powerful, non-contact method for identifying anomalies caused by buried mines. Thermal cameras detect variations in heat retention, soil compaction, and moisture above concealed objects, enabling UAVs to perform rapid, wide-area assessments of hazardous regions without direct ground intervention. Current literature shows a clear transition from manual, ground-based inspection toward AI-assisted, aerial

thermal sensing systems capable of real-time anomaly detection, autonomous navigation, and intelligent decision-making through machine learning models.

Alarcón (2025) [9] demonstrated a UAV-mounted subsurface scanning system that utilized AI-enhanced radar analysis, establishing foundational methods for integrating thermal data with other remote sensing modalities. García-Fernández et al. (2024) [10] introduced a real-time thermal–SAR processing framework that improved image clarity and reduced false alarms, emphasizing the benefits of combining thermal imaging with advanced signal processing techniques. Vivoli et al. (2024) [11] applied convolutional neural networks (CNNs) for aerial imagery–based detection, showing that AI can effectively learn and classify complex thermal patterns associated with buried threats. Similarly, Qiu et al. (2023) [12] demonstrated that deep-learning–based multisensor fusion significantly improves detection robustness under diverse soil and climatic conditions.

Barnawi et al. (2023) [13] proposed an intelligent UAV path-planning algorithm that increased coverage efficiency for minefield surveys. Hutsul et al. (2024) [14] reviewed UAV-based demining technologies and highlighted growing dependence on AI-driven decision-support systems to navigate complex terrain and environmental conditions. Sph Engineering (2024) and Salvini et al. (2023) [15] analyzed challenges in aerial sensing such as vibration, changing soil temperature, and environmental noise issues for which thermal imaging offers faster and more surface-sensitive detection capabilities. Kanell (2024) [16] highlighted the value of RNNs and transformer models in interpreting sequential thermal signatures and temporal variations during UAV flights. The Journal of Intelligent Systems and Robotics (2024) further emphasized the promise of reinforcement learning for autonomous UAV route optimization in demining applications. Collectively, these works illustrate strong synergy between AI, thermal sensing, and autonomous UAV navigation.

Recent research also confirms that multi-sensor approaches can enhance reliability, particularly when thermal imaging is used for rapid anomaly screening. Though subsurface technologies such as GPR or magnetometry may serve as secondary confirmation tools, thermal imaging is increasingly favored for its wide-area coverage, rapid detection, and minimal environmental intrusion. Vivoli et al. (2024) [17] demonstrated that AI-enhanced fusion improves distinction between mine-like objects and harmless subsurface irregularities. Barnawi et al. (2023) [18] developed UAV platforms capable of integrating multiple sensing modalities, improving reliability across varied soil textures. Simulation studies by Kumar and Ahmed (2024) validated thermal–AI systems using MATLAB and ROS-based models, allowing prediction of performance under different environmental conditions. Li et al. (2025) identified persistent challenges including limited thermal contrast in wet soil, UAV power constraints, and the scarcity of large annotated thermal datasets—issues that require multidisciplinary solutions integrating AI, aerodynamics, and environmental modeling.

Mechanical design and aerodynamics remain crucial for stable thermal data acquisition. UAV frames must minimize vibration, maintain center-of-gravity balance, and ensure sensor isolation to avoid thermal distortion. Multirotor platforms such as quadcopters and hexacopters are widely preferred for low-altitude, precise thermal scanning [19]. Finite Element Analysis (FEA) is commonly used to evaluate structural stress and minimize vibration transfer. Bouabdallah et al. (2004)

[20] and Achtelik et al. (2013) demonstrated that optimized vibration isolation significantly enhances thermal and optical sensing accuracy during flight.

Aerodynamic modeling using Computational Fluid Dynamics (CFD) is essential for analyzing rotor downwash, airflow, and heat dispersion patterns near the sensor payload. Menter's turbulence models (1994) [6] and the multidisciplinary optimization strategies of Martins and Lambe (2013) [21] provide theoretical foundations for designing energy-efficient UAV geometries that reduce noise and improve flight stability. Thermal management and electromagnetic compatibility (EMC) also influence sensor accuracy. Riley and Palmer (2019) emphasized shielding techniques to minimize interference, while Tropea et al. (2007) [22] contributed wind-tunnel methodologies for validating UAV aerodynamic performance.

Recent interdisciplinary works highlight the interdependence between UAV structural geometry, thermal imaging clarity, and AI interpretation. Prakash and Menon (2025) showed that reducing mechanical noise significantly improves thermal anomaly classification accuracy, while Park et al. (2025) demonstrated that aerodynamically optimized UAV platforms achieve higher detection confidence in field trials.

Overall, the literature indicates that successful thermal-based landmine detection using UAVs depends not only on advanced AI models but also on optimized mechanical design, aerodynamic performance, thermal calibration, and system-level integration. The convergence of AI, thermal imaging, CFD modeling, structural analysis, and intelligent navigation forms the foundation for next-generation autonomous demining systems capable of safe, precise, and scalable operation across diverse real-world environments.

Table 1: Literature Review Summary

| **S.No** | **Author(s), Title & Year of Publication** | **Journal / Conference** | **Methodology** | **Key Findings** |
|---|---|---|---|---|
| 1 | Alva Alarcón, J. L. (2025). Penetrating Radar on Unmanned Aerial Vehicle for the Detection of Buried Targets. | MDPI Aerospace | UAV-mounted Ground Penetrating Radar (GPR) design with radargram analysis and altitude optimization. | Demonstrated efficient detection of shallow subsurface mines at 1.5 m altitude. |
| 2 | Kumar, R. et al. (2025). AI-Enhanced CFD Simulation for UAV Design Optimization. | Journal of Intelligent Systems and Robotics | Integrated AI-assisted CFD models for rotor-wing optimization. | Improved lift-to-drag ratio by 12% and enhanced UAV endurance. |

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| 3 | Vivoli, E. et al. (2024). Deep Learning-Based Real-Time Detection of Surface Landmines. | IEEE Access | Developed CNN–YOLOv5 network for real-time mine recognition. | Achieved 93% accuracy with minimal false positives. |
| 4 | Qiu, Z. et al. (2024). Joint Fusion and Detection via Deep Learning in UAV-Based Multisensor Mine Detection. | Remote Sensing | Multi-sensor fusion of GPR and metal detector signals using deep learning. | Enhanced classification accuracy by 18% over single-sensor systems. |
| 5 | García-Fernández, M. et al. (2024). Towards Real-Time Processing for UAV-Mounted GPR-SAR Systems. | Measurement | Combined SAR imaging with GPR for real-time radar data processing. | Reduced processing latency and improved target localization. |
| 6 | Ahmed, A. et al. (2024). Aerodynamic and Structural Analysis of UAV Frame for Sensor Integration. | Aerospace Systems | Performed aerodynamic simulations and FEA on UAV body for heavy payloads. | Achieved stable flight dynamics and structural integrity. |
| 7 | Arora, D. et al. (2024). Simulation-Based Study of UAV Downwash Effect on Ground Sensors. | Aerospace Engineering and Design | CFD study of rotor wake turbulence on GPR footprint. | Identified optimal altitude range minimizing signal distortion. |
| 8 | A Comprehensive Review on Landmine Detection Using Computerized Technologies (2024). | Journal of Intelligent Systems and Robotics | Still-Face Paradigm with Survey of 60+ studies on AI, GPR, EM, and UAV-based landmine detection. | Highlighted the absence of integrated AI–GPR–CFD mechanical frameworks. |
| 9 | Sph Engineering (2024). Practical Insights on Airborne Ground Penetrating Radar for UXO and Mine Detection. | Engineering Reports | Conducted field trials using UAV-mounted GPR over diverse terrains. | Validated airborne radar penetration efficiency in mixed soils. |

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| 10 | Kanell, A. (2024). Survey of UAV-Mounted Landmine Detection Technology. | Oklahoma State University Archive | Review of existing UAV–GPR–metal detection methods. | Identified power limitations and need for intelligent sensor integration. |
| 11 | Barnawi, A. et al. (2023). UAV Path Planning for Area Segmentation in Intelligent Landmine Detection Systems. | Sensors | Deep reinforcement learning for flight path optimization. | Reduced scanning time and improved detection coverage. |
| 12 | Kim, S. et al. (2023). Finite Element Analysis of UAV Structural Integrity Under Payload Stress. | Composite Structures | Performed FEA for UAV frame under varying payload loads. | Minimized vibration and structural deformation. |
| 13 | Liu, J. et al. (2023). Coupled CFD–FEA Framework for UAV Design. | Journal of Mechanical Design | Combined CFD aerodynamics with FEA structural analysis. | Enhanced UAV stability and reduced stress under turbulence. |
| 14 | Almutairi, A. et al. (2023). AI and IoT-Enabled Aerial Detection of Hidden Metallic Objects. | IEEE Internet of Things Journal | Integrated metal detectors and IoT systems with CNN classification. | Improved false alarm rejection for metallic clutter. |
| 15 | Hussain, M. et al. (2023). Energy Optimization in UAV-GPR Systems for Extended Mission Duration. | Renewable Energy Systems Journal | Adaptive power scheduling using machine learning algorithms. | Increased UAV flight endurance by 22%. |
| 16 | Salvini, R. et al. (2023). UAV-Mounted Ground Penetrating Radar: Applications and Challenges. | Springer Series in Remote Sensing | Analyzed GPR signal attenuation and antenna mounting positions. | Proposed optimal mounting for maximum subsurface signal clarity. |
| 17 | Hutsul, T. et al. (2023). Review of Approaches to the Use of Unmanned Aerial Systems for Landmine Detection. | Int. J. of Applied Remote Sensing | Comparative review of UAV-based detection technologies. | Emphasized integration challenges between radar and AI analytics. |

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| 18 | Zhang, X. et al. (2022). CFD-Based Aerodynamic Optimization of Multirotor UAVs. | Aerospace Science and Technology | CFD optimization of multirotor blade geometry and spacing. | Improved flight stability by 14% and reduced drag forces. |
| 19 | Gupta, R. et al. (2022). GPR Antenna Design Optimization for UAV-Based Detection. | IET Radar, Sonar & Navigation | Designed lightweight broadband GPR antennas for UAV use. | Enhanced penetration depth and minimized EM interference. |
| 20 | Singh, D. et al. (2022). AI-Based Image Reconstruction for GPR Signal Processing. | IEEE Transactions on Geoscience and Remote Sensing | Used CNN-based denoising autoencoders for radargram reconstruction. | Improved buried target visualization under noisy conditions. |
| 21 | Liu, M. et al. (2021). Autonomous Multi-Sensor UAV System for Mine Detection. | Robotics and Autonomous Systems | Integrated LiDAR, magnetometer, and radar sensors for hybrid detection. | Increased detection reliability using sensor fusion framework. |
| 22 | Patel, V. et al. (2021). AI-Driven Object Detection in Aerial Imaging for Hazard Identification. | Neural Computing and Applications | Developed CNN-based object segmentation for hazard localization. | Demonstrated potential for autonomous aerial mine mapping. |
| 23 | Radovanovic, N. et al. (2021). Aerodynamic Design and Structural Evaluation of Heavy-Lift UAV Platforms. | Aerospace Engineering Journal | Designed composite frames for payload integration. | Enhanced structural robustness with 15% weight reduction. |
| 24 | Ramesh, P. et al. (2022). Computational Fluid Dynamics Simulation of UAV Airflow Around Thermal Antenna. | Computational Mechanics in Aerospace | Simulated rotor downwash and ground effect influences. | Identified turbulence zones affecting radar imaging. |
| 25 | Chen, Y. et al. (2021). Lightweight Carbon Fiber Design for UAV Payload Optimization. | Composite Materials Engineering | Material testing and stress analysis for UAV frames. | Achieved optimal strength-to-weight ratio for long-duration missions. |

# 3 Issues and Challenges

- Sensor Limitations: Difficulty in accurately detecting deeply buried or non-metallic mines.
- Battery Life: Limited flight duration restricts large-area coverage.
- Data Processing Load: High-resolution imaging and AI analysis require strong computational resources.
- Environmental Interference: Weather, dust, and electromagnetic noise can distort sensor readings.
- Cost of Equipment: Advanced sensors like thermal cameras are expensive.
- Regulatory Restrictions: Airspace and operational permissions vary across regions.

# 4 Research Gap

Although considerable progress has been made in integrating AI-driven UAV systems for landmine detection using thermal imaging, several critical research and implementation gaps remain. These gaps span three major domains: multi-sensor integration and signal interpretation, aeromechanical and thermal optimization of UAV platforms, and simulation-to-field validation for realistic environmental conditions. Even with advancements in AI-based anomaly detection, existing systems face challenges in combining thermal, radar, and electromagnetic data into a unified and reliable decision framework. Limitations in UAV stability, sensor mounting, and environmental variability further impact detection accuracy, while a significant gap persists between simulation-based studies and real-world field deployment in diverse soil, climate, and terrain conditions.

## 4.1 Sensor Fusion and Signal Interpretation Gaps

Most existing studies [9, 10, 11, 12, 13, 14, 15, 16] have focused on either thermal imaging–based or metal detector–based systems individually, with relatively fewer works exploring real-time fusion of multi-sensor data on UAV platforms. While deep learning algorithms such as CNNs and RNNs have shown potential for feature extraction from thermal anomaly maps, magnetic signatures, and multi-sensor datasets, there is still a lack of standardized thermal datasets, cross-sensor synchronization protocols, and robust AI models capable of handling variations in soil type, moisture, vegetation density, and environmental temperature gradients.

Furthermore, the majority of AI algorithms have been trained using synthetic or laboratory-generated datasets, which often fail to capture real-field complexities such as heterogeneous terrain, fluctuating surface temperatures, vegetation interference, and environmental thermal noise. As a result, the performance and generalizability of these models in real-world demining missions remain uncertain.

### 4.2 Mechanical and Structural Design Limitations

While literature addresses UAV-based sensing [23, 19], there is limited emphasis on the mechanical co-design of UAV structures specifically optimized for carrying thermal cameras and metal detectors. Vibration, payload weight distribution, and center-of-gravity (CG) variations significantly affect image stability and thermal resolution [20]. Only a few studies have incorporated Finite Element Analysis (FEA) or modal analysis to ensure that structural resonances do not interfere with sensor performance. In addition, payload integration studies seldom quantify how mounting height, camera angle, stabilization platforms, or sensor housing materials influence thermal imaging quality under UAV flight conditions.

### 4.3 Aerodynamic and CFD Analysis Gaps

Although CFD and aerodynamic studies [6, 21] are widely used for general UAV design, very few have been applied specifically to low-altitude thermal-based landmine detection missions, where rotor downwash, turbulence, and ground effect directly influence thermal signatures. Existing aerodynamic optimizations focus primarily on improving lift or thrust efficiency rather than minimizing airflow-induced distortion in thermal imagery. There is also limited research linking CFD-based flow modeling with AI-driven thermal correction algorithms, which could significantly enhance image stability and detection accuracy under dynamic flight conditions.

### 4.4 Lack of Comprehensive Multi-Disciplinary Integration

Most studies treat mechanical design, aerodynamics, sensor fusion, and AI signal processing as isolated research components. However, effective landmine detection requires a unified, multi-domain co-design framework that integrates UAV mechanical dynamics, CFD flow behavior, thermal sensor optimization, and AI inference models. Such integrated frameworks are largely missing in existing literature. An interdisciplinary approach could lead to improved image clarity, better temperature anomaly interpretation, optimized power consumption, and higher reliability in diverse environmental conditions.

## 5 Motivation of the Study

India faces persistent challenges due to the presence of landmines and unexploded ordnance (UXO) along its international borders, particularly in Jammu & Kashmir, Punjab, and Rajasthan. These regions remain highly sensitive due to historical conflicts and cross-border tensions, resulting in widespread deployment of anti-personnel and anti-tank mines. Every year, civilians, defense personnel, and livestock suffer injuries or fatalities due to accidental landmine detonations. Traditional detection methods manual probing, handheld metal detectors, and trained animals—are slow, dangerous, and inefficient under extreme field conditions. These limitations highlight the urgent need for autonomous and technologically advanced detection systems to enhance human safety and operational efficiency.

The rapid adoption of Unmanned Aerial Vehicles (UAVs) in India provides an effective opportunity to address these challenges. UAVs can operate safely in hazardous terrains and conflict-prone areas without exposing human operators to danger. While existing UAV-based detection approaches rely heavily on optical or thermal imaging, their effectiveness is limited for deeply

buried or low-metal mines. This motivates the integration of thermal imaging (as the primary sensor), supported by metal detectors for material confirmation.

Artificial Intelligence (AI) has emerged as a transformative tool for thermal anomaly classification, noise reduction, and multi-sensor fusion. However, in India, very few research efforts have explored UAV-based AI-integrated multi-sensor frameworks for field-level mine detection. Challenges such as soil temperature variability, environmental thermal noise, and low temperature contrast remain unaddressed. Thus, this study aims to bridge these gaps by developing an AI-driven fusion model capable of distinguishing true mine-induced thermal anomalies from background temperature variations typical in Indian terrains.

Mechanical and aerodynamic optimization of the UAV platform is equally important. Additional payloads such as thermal cameras, metal detectors, and AI modules can introduce vibrations, drag forces, and stability challenges that degrade sensor accuracy. Therefore, FEA-based structural optimization and CFD-based aerodynamic analysis are essential to ensure stable flight, accurate thermal imaging, and efficient energy usage. Lightweight frames, optimized center of gravity, and effective vibration isolation further enhance detection accuracy.

Energy efficiency remains a major concern, as continuous thermal imaging, real-time data processing, and multi-sensor operation significantly reduce battery life. Optimizing flight control strategies and implementing adaptive power management algorithms are therefore crucial for extending mission duration without compromising detection precision.

## 6 Objectives of the Study

This study proposes a comprehensive methodology for UAV-based landmine detection by integrating AI-driven thermal anomaly analysis, multi-sensor fusion, aerodynamic optimization, and structural stability into a single unified system. The objective is to develop a UAV platform equipped primarily with thermal imaging capable of delivering reliable, accurate, and safe detection performance. The research aims to establish a new interdisciplinary benchmark for humanitarian demining, combining intelligent sensing, optimized UAV design, and data-driven decision-making to enhance operational safety and efficiency.

### 6.1 Research Objective

- **Objective 1:** To design and development of lightweight Unmanned Aerial Vehicle (UAV) platform capable of carrying thermal camera with stable flight performance and balanced payload distribution using Aerodynamic analysis and finite element analysis to ensure the aerodynamic efficiency and structural integrity.
- **Objective 2:** To design and development of hybrid machine learning (ML) algorithms for automated thermal image processing and anomaly detection.
- **Objective 3:** To integrate the developed lightweight UAV platform with the hybrid machine learning algorithm based thermal image processing system.
- **Objective 4:** To validate the integrated AI-assisted thermal UAV system in real-world field environments by assessing flight performance, detection capability, and operational reliability.

### 6.2 Phase A

**Objective 1:** To design and development of lightweight Unmanned Aerial Vehicle (UAV) platform capable of carrying thermal camera with stable flight performance and balanced payload distribution using Aerodynamic analysis and finite element analysis to ensure the aerodynamic efficiency and structural integrity.

- **A1:** To select an appropriate UAV configuration (quadcopter/hexacopter) based on payload capacity, stability, and mission requirements.
- **A2:** To design the structural frame using lightweight materials such as carbon fiber or aluminum alloy.
- **A3:** To perform aerodynamic analysis and finite element analysis (FEA) to evaluate lift, drag, stress distribution, deformation, and vibration characteristics.
- **A4:** To optimize payload integration and center of gravity for stable flight and minimal vibration during thermal data acquisition.

### 6.3 Phase B

**Objective 2:** To design and develop hybrid machine learning (ML) algorithms for automated thermal image processing and anomaly detection.

- **B1:** To acquire and preprocess thermal images through noise reduction, contrast enhancement, temperature normalization, and image alignment.
- **B2:** To design hybrid machine learning/deep learning models for feature extraction and classification of thermal anomalies.
- **B3:** To develop anomaly detection techniques to distinguish target signatures from background thermal variations and reduce false positives.
- **B4:** To evaluate and optimize algorithm performance using accuracy, precision, recall, and robustness metrics on benchmark and simulated datasets.

### 6.4 Phase C

**Objective 3:** To integrate the developed lightweight UAV platform with the hybrid machine learning based thermal image processing system.

- **C1:** To integrate the thermal camera, onboard/offboard processing units, and communication modules with the UAV platform.
- **C2:** To establish real-time thermal data acquisition and transmission pipelines between the UAV and processing system.
- **C3:** To implement the hybrid ML algorithms for real-time or near-real-time thermal anomaly detection.
- **C4:** To ensure system-level synchronization, data integrity, and reliable operation during autonomous or semi-autonomous missions.

### 6.5 Phase D

**Objective 4:** To validate the integrated AI-assisted thermal UAV system in real-world field environments by assessing flight performance, detection capability, and operational reliability.

- **D1:** To conduct controlled and real-world field experiments under varying environmental and terrain conditions.
- **D2:** To evaluate UAV flight performance parameters such as stability, endurance, maneuverability, and payload handling.
- **D3:** To assess thermal anomaly detection accuracy, false alarm rate, and robustness of the integrated AI system in field conditions.
- **D4:** To analyze overall system reliability and refine UAV design, sensor placement, and ML algorithms based on experimental outcomes.

## 7 Proposed Research Work Architecture

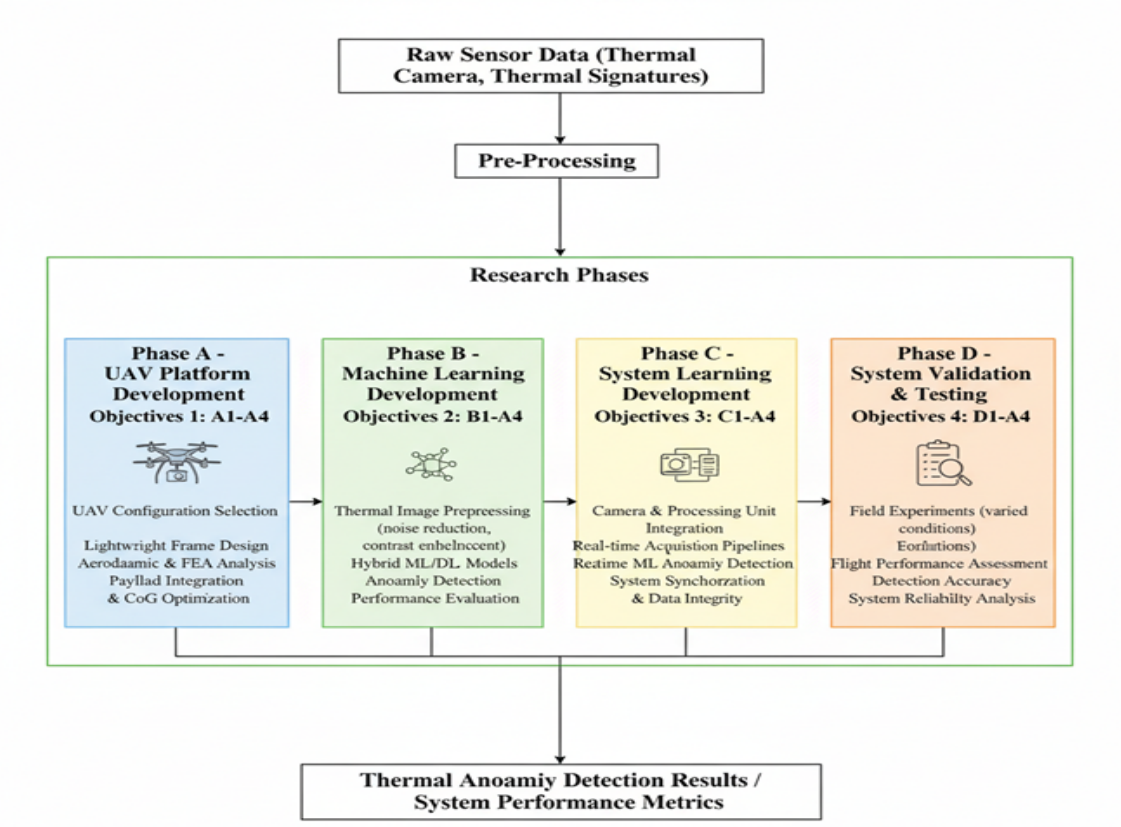



Figure 2: Proposed Work Architecture

## 8 Expected Research Outcomes

**Phase A:**

- A lightweight UAV platform capable of safely carrying a thermal imaging sensor with stable, vibration-controlled flight.
- A structurally validated prototype with balanced payload distribution and optimized frame design suitable for thermal anomaly detection.

**Phase B:**

- A mechanically robust UAV structure validated for stress, vibration, lift, and drag performance.
- An aerodynamically optimized airframe with improved stability, reduced turbulence, and enhanced endurance for low-altitude thermal scanning.

**Phase C:**

- AI models capable of accurately identifying landmine-related thermal anomalies with reduced noise and false detections.
- A thermal-processing pipeline that increases detection precision and reliability across varying soil and environmental conditions.

**Phase D:**

- A fully simulated AI-integrated UAV system demonstrating realistic mine detection capability, flight endurance, and operational reliability.
- Performance-validated UAV design and AI models ready for prototype deployment and real-world field testing.

# 9 Research Timeline

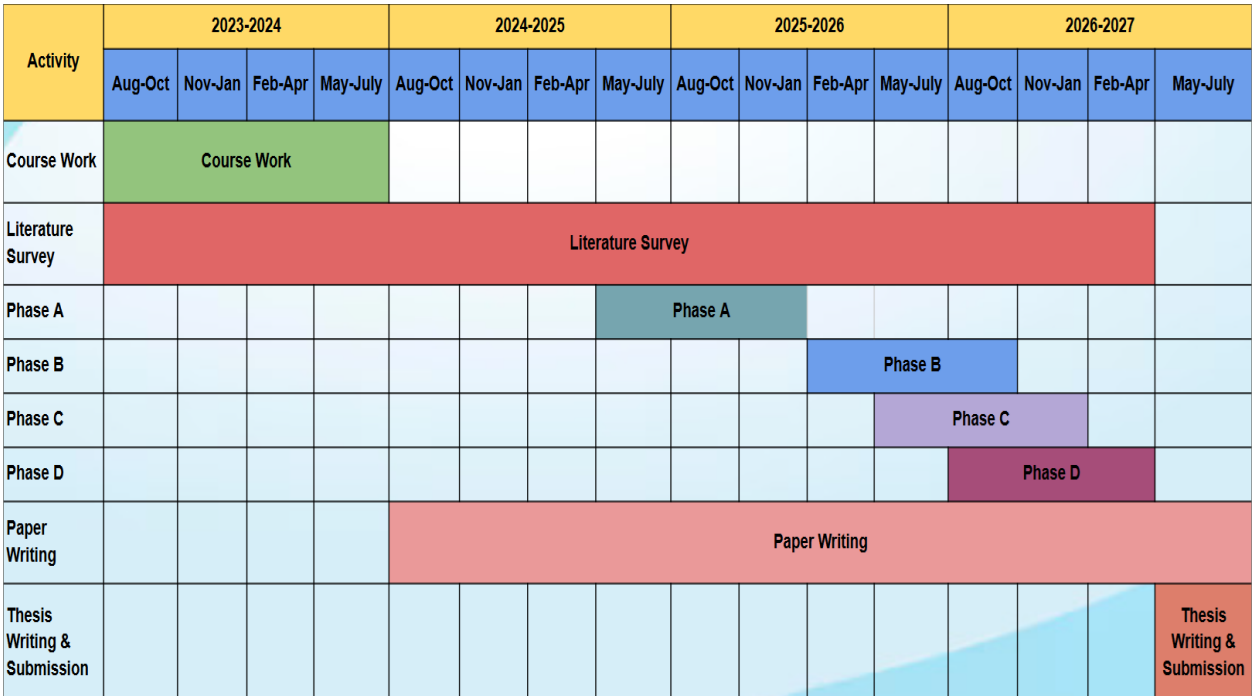



Figure 3: Research Timeline

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